EE341 — Quiz 1

Condensed revision sheet + problem set with full solutions

Communication Systems I, Autumn 2026 — covering everything through the superheterodyne receiver

Weighted towards the three areas you flagged: **spectra of modulated signals**, **modulator and demodulator circuits**, and **bandpass / complex-envelope / Hilbert**. Part I is the reference sheet — read it once. Part II is 14 problems; **do them closed-book, then mark**. The instructor said quizzes reuse homework problems with minor modifications, so the format matters as much as the content.

Conventions used throughout: ordinary frequency f in Hz; message $x(t)$ real, zero-mean, bandlimited to W, normalised so $\max|x| = 1$; carrier $f_c \gg W$; $x_{bp} = \mathrm{Re}\{x_\ell e^{j2\pi f_c t}\}$.

Contents

PART I — Revision sheet

1 The four AM schemes, side by side

Scheme	$x_c(t)$	BW	Complex env. x_ℓ	Detector
DSB-SC	$A\,x(t)\cos 2\pi f_c t$	$2W$	$A\,x(t)$ (real, sign-changing)	coherent only
DSB-AM	$A\big(1+\mu x(t)\big)\cos 2\pi f_c t$	$2W$	$A(1+\mu x)$ (>0 if $\mu \le 1$)	envelope (or coherent)
SSB	$\frac{A}{2}\big[x\cos 2\pi f_c t \mp \hat{x}\sin 2\pi f_c t\big]$	W	$\frac{A}{2}(x \pm j\hat{x})$	coherent only
VSB	partial vestige of one sideband	$\approx W + f_v$	—	coherent (or envelope if carrier added)

In SSB the upper sign ($-\hat{x}\sin$) gives the **upper** sideband, the lower sign the **lower** sideband. Mnemonic: *USB = minus.*

Power (with $\langle x \rangle = 0$, $P_x = \langle x^2 \rangle$).

$$P_{\text{DSB-SC}} = \frac{A^2}{2} P_x, \qquad P_{\text{DSB-AM}} = \underbrace{\frac{A^2}{2}}_{\text{carrier}} + \underbrace{\frac{A^2\mu^2}{2} P_x}_{\text{sidebands}}, \qquad \eta = \frac{\mu^2 P_x}{1+\mu^2 P_x}.$$

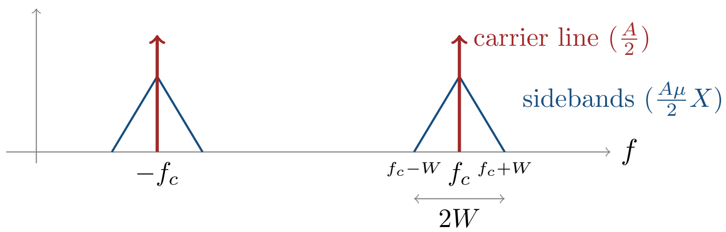
Ceiling: $\eta = 1/3$ for a sinusoidal message at $\mu = 1$; $\eta = 1/2$ for a square-wave message. Always: $\eta_{\text{DSB-SC}} = 100\%$.

2 Spectra — the recipe

Every spectrum question in this course is the **modulation property** applied once or twice:

$$x(t)\cos 2\pi f_c t \overset{\mathcal{F}}{\longleftrightarrow} \tfrac{1}{2}X(f - f_c) + \tfrac{1}{2}X(f + f_c).$$

Procedure. (i) Write $x_c(t)$ as a sum of terms. (ii) Transform each: a constant $\to$ impulse; a cosine $\to$ two impulses; $x(t)\cos \to$ two half-height copies of X. (iii) Draw on one axis, labelling $\pm f_c$, band edges $f_c \pm W$, and *heights*. (iv) Read off bandwidth = extent of one positive-frequency group.



Above: **DSB-AM**. Remove the two arrows and it is **DSB-SC**. Keep only the outer half of each triangle and it is **USB**; the inner half, **LSB**.

Four spectrum traps.

1. **Heights halve.** Modulation gives $\frac{1}{2}X(f \mp f_c)$, not X. With amplitude A in front, $\frac{A}{2}$.
2. **The carrier is an impulse**, drawn as an arrow; its label is its *weight* (area), here $\frac{A}{2}$. It is not "infinitely tall signal".
3. **Sideband inversion.** In the *lower* sideband (and on the negative-frequency side) the message spectrum appears **mirrored**. Draw an asymmetric $X(f)$ to show you know this.
4. **Bandwidth of the real signal** is the extent of the positive-frequency group ($2W$ for DSB), *not* the total occupancy including negative frequencies.

3 Modulator circuits

All three do the same job — multiply — and differ only in how they cheat.

(a) Ring modulator (double-balanced). Carrier square wave switches an opposite pair of diodes each half-cycle; one pair passes $x(t)$ straight, the other with leads swapped.

$$v_{\text{out}} = x(t)c(t), \qquad c(t) = \frac{4}{\pi}\left[\cos 2\pi f_c t - \tfrac{1}{3}\cos 6\pi f_c t + \tfrac{1}{5}\cos 10\pi f_c t - \cdots\right]$$

BPF at f_c, bandwidth $2W \Rightarrow \boxed{v = \frac{4}{\pi}x(t)\cos 2\pi f_c t}$, i.e. **DSB-SC**. Requires $A_c \gg |x|$ (switching set by carrier alone) and $f_c > W$ (copy at f_c clear of the one at $3f_c$). Suppressed carrier because $c(t)$ has **no DC term**.

(b) Switching modulator (single-balanced). Diode switched by carrier; input is $x(t)+A\cos 2\pi f_c t$, switching function unipolar: $s(t) = \frac{1}{2} + \frac{2}{\pi}\cos 2\pi f_c t - \cdots$. The DC $\frac{1}{2}$ lets the *carrier* through:

$$v_{\mathrm{BPF}} = \tfrac{A}{2}\cos 2\pi f_c t + \tfrac{2}{\pi}x(t)\cos 2\pi f_c t \quad \Rightarrow \quad \textbf{DSB-AM}.$$

(c) Square-law modulator. Device $v_o = a_1 v_i + a_2 v_i^2$ with $v_i = x(t) + \cos 2\pi f_c t$:

$$v_o = \underbrace{a_1 x + a_2 x^2}_{\text{baseband, } 2W} + \underbrace{\tfrac{a_2}{2}}_{\text{DC}} + \underbrace{a_1\left(1 + \tfrac{2a_2}{a_1}x\right)\cos 2\pi f_c t}_{\text{DSB-AM, } \mu=2a_2/a_1} + \underbrace{\tfrac{a_2}{2}\cos 4\pi f_c t}_{2f_c}.$$

BPF at f_c gives DSB-AM with $\mu = 2a_2/a_1$. **Condition** $f_c > 3W$ so the baseband group (width $2W$) does not reach $f_c - W$.

4 Demodulator circuits

Coherent (synchronous / product). Multiply by a local $\cos(2\pi f_c t + \phi)$, lowpass:

$$\boxed{y(t) = \tfrac{A}{2}x(t)\cos\phi}$$

- $\phi = 90^\circ \Rightarrow$ output **zero** (quadrature null). Needs carrier recovery: PLL, Costas loop, squaring loop, or a pilot tone.
- Frequency error Δf: $y = \frac{A}{2}x(t)\cos(2\pi\Delta f\, t + \phi)$ — slow beating.
- Works for DSB-SC, DSB-AM, SSB, VSB. The only universal detector.

Envelope detector. Diode + RC. Output $= \left|A(1+\mu x)\right|$.
- Works **iff** $1 + \mu x \geq 0$ for all t, i.e. $\mu \leq 1$. Over-modulation folds the envelope and destroys the sign irrecoverably.
- Fails outright for DSB-SC (x_ℓ real and sign-changing — "permanently over-modulated").
- No phase reference needed — that is what makes it cheap, and why broadcast AM wastes power on a carrier.

Time-constant design.

$$\frac{1}{f_c} \ll RC \ll \frac{1}{W}, \qquad \text{precisely} \qquad RC \leq \frac{\sqrt{1-\mu^2}}{\mu \cdot 2\pi W}.$$

Too small $\Rightarrow$ ripple at f_c. Too large $\Rightarrow$ **diagonal clipping** (failure-to-follow). Note it tightens as $\mu \to 1$.

5 Superheterodyne receiver

Tuning a high-Q bandpass filter across the whole band is impractical, so instead **shift every station to one fixed intermediate frequency** and do the hard filtering there once.

$$f_{LO} = f_c + f_{IF} \text{ (high-side)}, \qquad \boxed{f_{\text{image}} = f_c + 2f_{IF}}$$

Both f_c and f_{image} mix down to f_{IF} (the mixer only sees $|f - f_{LO}|$), so the image must be killed **before** the mixer by the RF preselector. AM broadcast: band 540–1600 kHz, $f_{IF} = 455$ kHz.

Why f_{IF} is a compromise. Large f_{IF} pushes the image far away (easy to reject) but makes the fixed IF filter harder to build selectively. Small f_{IF} gives good adjacent-channel selectivity but a close-in image. 455 kHz is the historical compromise for the AM band.

6 Bandpass signals, complex envelope, Hilbert

$$\hat{x}(t) = x(t) * \frac{1}{\pi t}, \qquad H(f) = -\mathrm{j}\,\mathrm{sgn}(f) \quad (\text{a } -90^\circ \text{ phase shifter, } |H| = 1)$$

$$x_+(t) = \tfrac{1}{2}\big(x_{bp} + \mathrm{j}\hat{x}_{bp}\big), \qquad x_\ell(t) = 2x_+(t)e^{-\mathrm{j}2\pi f_c t} = x_I + \mathrm{j}x_Q,$$

$$\boxed{x_{bp}(t) = \mathrm{Re}\{x_\ell e^{\mathrm{j}2\pi f_c t}\} = x_I \cos 2\pi f_c t - x_Q \sin 2\pi f_c t}$$

$$x_I = x_{bp} \cos 2\pi f_c t + \hat{x}_{bp} \sin 2\pi f_c t, \qquad x_Q = \hat{x}_{bp} \cos 2\pi f_c t - x_{bp} \sin 2\pi f_c t,$$

$$A(t) = \sqrt{x_I^2 + x_Q^2} = |x_\ell|, \qquad \phi(t) = \arctan \frac{x_Q}{x_I}.$$

Hilbert facts: $\widehat{\cos} = \sin$, $\widehat{\sin} = -\cos$, $\hat{\hat{x}} = -x$, $\widehat{e^{\mathrm{j}2\pi f_0 t}} = -\mathrm{j}e^{\mathrm{j}2\pi f_0 t}$ $(f_0 > 0)$. Amplitudes never change.

Finding x_ℓ — the read-off method. Never integrate. Force the signal into the form $\mathrm{Re}\{(\cdots)e^{\mathrm{j}2\pi f_c t}\}$ and read off the bracket. Check: for $x_{bp} = A\cos 2\pi f_c t$ you must get $x_\ell = A$.

Two properties that do most of the work:

1. **Envelopes add.** Bandpass signals sharing f_c have complex envelopes that superpose (Re is linear).
2. **Filtering maps to filtering.** If $h_{bp} = 2\,\mathrm{Re}\{h_\ell e^{\mathrm{j}2\pi f_c t}\}$ then $y_\ell = x_\ell * h_\ell$. Hence a whole link can be simulated at baseband at a rate set by W, not f_c.

The five that cost marks.

1. The minus sign in $x_{bp} = x_I \cos - x_Q \sin$.
2. Dropping the $4/\pi$ (ring) or $2/\pi$ (switching) gain — fatal in power questions.
3. Forgetting to state $f_c > W$ (or $f_c > 3W$ for square-law) when a filter must separate copies.
4. Using $|X(f)|^2$ for a modulated carrier — it is a *power* signal; use PSD.
5. Mixing the $\frac{1}{2}/2/\sqrt{2}$ conventions between x_+, x_ℓ and different textbooks.

PART II — Problem set

Do these closed-book first. Suggested timing: A (spectra) 25 min, B (circuits) 35 min, C (bandpass) 35 min, D (power) 15 min. Solutions begin on the page after each group — resist.

Group A — Spectra of modulated signals

Problem 1. A message has $X(f)$ triangular: $X(f) = 1 - |f|/W$ for $|f| \le W$, zero outside. It is transmitted as $x_c(t) = A\big(1 + \mu x(t)\big)\cos 2\pi f_c t$ with $f_c \gg W$.
(a) Write $X_c(f)$ and sketch it, labelling every height and frequency.
(b) State the transmitted bandwidth.
(c) What fraction of the transmitted power is in the sidebands if $\mu = 0.6$ and $P_x = 0.2$?

Solution. (a) Split the signal first: $x_c = A\cos 2\pi f_c t + A\mu x(t)\cos 2\pi f_c t$. Transform each piece.

$$X_c(f) = \underbrace{\frac{A}{2}\big[\delta(f-f_c) + \delta(f+f_c)\big]}_{\text{carrier}} + \underbrace{\frac{A\mu}{2}\big[X(f-f_c) + X(f+f_c)\big]}_{\text{sidebands}}.$$

Sketch: at $+f_c$ an arrow of *weight* $A/2$, plus a triangle of peak height $A\mu/2$ spanning $f_c - W$ to $f_c + W$; mirror image at $-f_c$. Since X is even here, the two sidebands look alike — say so, and note that for asymmetric X the lower sideband would be mirrored.

(b) $B = 2W$ (extent of the positive-frequency group).

(c) $\eta = \dfrac{\mu^2 P_x}{1 + \mu^2 P_x} = \dfrac{0.36 \times 0.2}{1 + 0.36 \times 0.2} = \dfrac{0.072}{1.072} = 6.7\%.$

Problem 2. A square-law device $v_o = a_1 v_i + a_2 v_i^2$ is driven by $v_i = x(t) + \cos 2\pi f_c t$, with x bandlimited to W. List **every** spectral group at the output with its centre frequency and width, and state the condition on f_c for a bandpass filter to extract a clean AM signal. What is the resulting modulation index?

Solution. Expand:

$$v_o = a_1 x + a_1 \cos 2\pi f_c t + a_2 x^2 + 2a_2 x \cos 2\pi f_c t + a_2 \cos^2 2\pi f_c t, \qquad \cos^2 2\pi f_c t = \tfrac{1}{2} + \tfrac{1}{2}\cos 4\pi f_c t.$$

Grouping:

Term	**Centre**	**Width**	**Occupies**
$a_2/2$	DC	—	$f = 0$ (impulse)
$a_1 x + a_2 x^2$	0	$2W$ (x^2 doubles it)	$[-2W,\, 2W]$
$\big(a_1 + 2a_2 x\big)\cos 2\pi f_c t$	$\pm f_c$	$2W$	$[f_c - W,\, f_c + W]$
$\frac{a_2}{2}\cos 4\pi f_c t$	$\pm 2f_c$	—	impulse at $2f_c$

The wanted term is $a_1\big(1 + \frac{2a_2}{a_1}x\big)\cos 2\pi f_c t$, i.e. DSB-AM with

$$\boxed{\mu = \frac{2a_2}{a_1}}.$$

Condition. The baseband group reaches up to $2W$; the wanted band starts at $f_c - W$. Need

$$f_c - W > 2W \quad\Longrightarrow\quad \boxed{f_c > 3W}.$$

(The $2f_c$ impulse is far above and never a problem.) Note also $\mu \le 1$ demands $2a_2 \le a_1$ — the message must be kept small relative to the carrier, the same condition as always.

Problem 3. Sketch and compare the magnitude spectra of (i) DSB-SC, (ii) USB-SSB and (iii) LSB-SSB for the same message, and state each bandwidth. Then explain in one sentence why SSB halves the bandwidth without losing information.

Solution. (i) DSB-SC: two triangles, centred $\pm f_c$, each spanning $2W$, height $\frac{A}{2}$. $B = 2W$.
(ii) USB: keep only the parts *outside* $\pm f_c$, i.e. $[f_c, f_c + W]$ and $[-f_c - W, -f_c]$. $B = W$.
(iii) LSB: keep only the parts *inside*: $[f_c - W, f_c]$ and $[-f_c, -f_c + W]$. $B = W$.

Why nothing is lost: $x(t)$ is real, so $X(f)$ has Hermitian symmetry and the two sidebands are mirror images carrying identical information — transmitting one is enough, and the receiver reconstructs the other.

Group B — Modulator and demodulator circuits

Problem 4. For the ring modulator, show that the output after a bandpass filter centred at f_c with bandwidth $2W$ is $\frac{4}{\pi}x(t)\cos 2\pi f_c t$. State the two conditions required and explain why the carrier is suppressed.

Solution. The carrier square wave switches one opposite pair of diodes on per half-cycle: one pair connects the message transformer to the output transformer straight through, the other with the leads swapped. So the output is the message multiplied by a ± 1 square wave: $v_{\text{out}}(t) = x(t)\,c(t)$.

Fourier series of $c(t)$ (half-wave symmetric $\Rightarrow$ odd harmonics only; even about the chosen origin $\Rightarrow$ cosines):

$$c(t) = \frac{4}{\pi}\cos 2\pi f_c t - \frac{4}{3\pi}\cos 6\pi f_c t + \frac{4}{5\pi}\cos 10\pi f_c t - \cdots$$

Multiplying term by term places a copy of X around $f_c, 3f_c, 5f_c, \ldots$ with amplitudes $\frac{4}{\pi}, \frac{4}{3\pi}, \ldots$ The BPF keeps the first:

$$\boxed{v_{\text{BPF}}(t) = \frac{4}{\pi}x(t)\cos 2\pi f_c t} \qquad \text{(DSB-SC, gain } 4/\pi \approx 1.27\text{)}.$$

Conditions. (i) $A_c \gg |x(t)|$ so the diode switching is set by the carrier alone and $c(t)$ is a clean square wave — otherwise the switching becomes message-dependent and the product is no longer linear in x. (ii) $f_c + W < 3f_c - W \Rightarrow f_c > W$, so the wanted copy is separable from the one at $3f_c$.

Carrier suppression. $c(t)$ spends equal time at $+1$ and -1, so it has **zero DC component**; with no DC there is nothing to let a bare carrier term through. Structurally, the circuit is *double-balanced*: the carrier is injected at the centre tap of the output transformer, driving both halves of the winding equally and oppositely, so it cancels in the secondary.

Problem 5. A DSB-SC signal $x_c(t) = Ax(t)\cos 2\pi f_c t$ is coherently detected with a local oscillator $\cos(2\pi(f_c + \Delta f)t + \phi)$.
(a) Find the lowpass-filtered output.
(b) What happens at $\phi = 90^\circ, \Delta f = 0$?
(c) If $\Delta f = 100$ Hz and the message is speech, describe what you hear.

Solution. (a) Product-to-sum:

$$Ax\cos 2\pi f_c t \cdot \cos(2\pi(f_c + \Delta f)t + \phi) = \frac{Ax}{2}\Big[\cos(2\pi\Delta f\,t + \phi) + \cos(2\pi(2f_c + \Delta f)t + \phi)\Big].$$

LPF removes the $2f_c$ term:

$$\boxed{y(t) = \frac{A}{2}x(t)\cos(2\pi\Delta f\,t + \phi)}.$$

(b) $\Delta f = 0, \phi = 90^\circ$: $y = 0$. Total loss of signal — the **quadrature null**. This is why DSB-SC needs carrier recovery.

(c) The output is the speech multiplied by a 100 Hz cosine: the audio **fades in and out 200 times a second** (twice per cycle of the beat, since it passes through zero twice), with a sign reversal each time. Unintelligible — a warbling/garbled sound. Note the fading is at $2\Delta f$ in perceived rate.

Problem 6. Design the envelope detector for a broadcast AM receiver: $f_c = 1$ MHz, $W = 5$ kHz, maximum modulation index $\mu = 0.8$. Find the permitted range of RC and choose a value. What happens at $RC = 1$ ms?

Solution. **Lower bound** (must not discharge between carrier peaks):

$$RC \gg \frac{1}{f_c} = 1\ \mu\text{s}.$$

Upper bound (must follow the envelope down):

$$RC \le \frac{\sqrt{1-\mu^2}}{\mu\, 2\pi W} = \frac{\sqrt{1-0.64}}{0.8 \times 2\pi \times 5000} = \frac{0.6}{25133} = 2.39 \times 10^{-5}\ \text{s} = 23.9\ \mu\text{s}.$$

So $1\ \mu\text{s} \ll RC \le 23.9\ \mu\text{s}$. A sensible choice is $RC \approx 10\ \mu\text{s}$ — ten carrier periods (good smoothing) and well inside the upper limit.

At $RC = 1$ ms: forty times too large. The capacitor cannot discharge fast enough on falling envelope edges, so the output cuts straight-line corners — **diagonal clipping** (failure-to-follow distortion), and the recovered audio is badly distorted on loud passages.

Problem 7. A superheterodyne receiver has $f_{IF} = 455$ kHz and high-side local oscillator. It is tuned to a station at 600 kHz.

(a) Find f_{LO} and the image frequency.
(b) The AM band is 540–1600 kHz. Is the image inside the band? Why is that a problem?
(c) What suppresses the image, and why can it be a low-Q filter?

Solution. (a) $f_{LO} = f_c + f_{IF} = 600 + 455 = 1055$ kHz.

$$f_{\text{image}} = f_c + 2f_{IF} = 600 + 910 = \boxed{1510\text{ kHz}}.$$

Check: $|1510 - 1055| = 455\text{ kHz} = f_{IF}$. ✓Both stations mix down to the same IF because the mixer responds to $|f - f_{LO}|$ and cannot tell which side of f_{LO} the input came from.

(b) **Yes** — 1510 kHz is inside 540–1600 kHz, so a real broadcast station can sit exactly there. Without rejection you would hear both stations superimposed, with no way to separate them after the mixer: once both are at 455 kHz, the IF filter cannot distinguish them.

(c) The **RF preselector** before the mixer. It needs only enough selectivity to reject something $2f_{IF} = 910$ kHz away — a wide, gentle, low-Q tuned circuit suffices. All the *sharp* filtering (separating 10 kHz-spaced adjacent channels) is done at the fixed IF, where a high-Q filter can be built once and never retuned. That division of labour is the entire point of the superheterodyne architecture.

Group C — Bandpass signals, complex envelope, Hilbert

Problem 8. Find x_ℓ, x_I, x_Q, $A(t)$ and $\phi(t)$ with respect to carrier f_c for:

(a) $x_{bp}(t) = A\cos(2\pi f_c t + \theta_0)$
(b) $x_{bp}(t) = m(t)\cos 2\pi f_c t - n(t)\sin 2\pi f_c t$, m, n real lowpass
(c) $x_{bp}(t) = A\cos(2\pi(f_c + \Delta f)t)$

Solution. Use the read-off method every time: force into $\text{Re}\{(\cdots)e^{j2\pi f_c t}\}$.

(a) $A\cos(2\pi f_c t + \theta_0) = \text{Re}\{Ae^{j\theta_0}e^{j2\pi f_c t}\}$, so $x_\ell = Ae^{j\theta_0}$ (*a constant*). Hence $x_I = A\cos\theta_0$, $x_Q = A\sin\theta_0$, $A(t) = A$, $\phi = \theta_0$. A phase offset parks the phasor at a fixed angle.

(b) This is already in the canonical form $x_I \cos - x_Q \sin$, so read off directly: $x_\ell = m + jn$, $x_I = m$, $x_Q = n$, $A = \sqrt{m^2+n^2}$, $\phi = \arctan(n/m)$. *Watch the minus sign* — had the signal been $m\cos + n\sin$, then $x_Q = -n$.

(c) $\text{Re}\{e^{j2\pi\Delta f t} \cdot e^{j2\pi f_c t}\} \cdot A$, so $x_\ell = Ae^{j2\pi\Delta f t}$: $x_I = A\cos 2\pi\Delta f t$, $x_Q = A\sin 2\pi\Delta f t$, $A(t) = A$ (constant), $\phi(t) = 2\pi\Delta f t$. **A frequency offset is a phasor rotating at Δf**; all the information is in the angle, none in the envelope.

Problem 9. Starting from $H(f) = -\mathrm{j}\,\mathrm{sgn}(f)$, derive $\widehat{\cos 2\pi f_c t}$ and $\widehat{\sin 2\pi f_c t}$, and hence show $\hat{\hat{x}} = -x$ for these signals.

Solution. $\cos 2\pi f_c t \xleftrightarrow{\mathcal{F}} \frac{1}{2}\delta(f - f_c) + \frac{1}{2}\delta(f + f_c)$. Multiply each impulse by H at that frequency (sampling property):

$$\text{at } +f_c: \; H = -\mathrm{j} \Rightarrow -\tfrac{\mathrm{j}}{2}\delta(f - f_c); \qquad \text{at } -f_c: \; H = +\mathrm{j} \Rightarrow +\tfrac{\mathrm{j}}{2}\delta(f + f_c).$$

Compare $\sin 2\pi f_c t \xleftrightarrow{\mathcal{F}} \frac{1}{2\mathrm{j}}\delta(f-f_c) - \frac{1}{2\mathrm{j}}\delta(f+f_c) = -\frac{\mathrm{j}}{2}\delta(f-f_c) + \frac{\mathrm{j}}{2}\delta(f+f_c)$. Identical, so $\boxed{\widehat{\cos} = \sin}$.

Same computation on sin gives $\widehat{\sin} = -\cos$. Applying twice: $\cos \to \sin \to -\cos$ and $\sin \to -\cos \to -\sin$, i.e. $\hat{\hat{x}} = -x$. ✓(Consistent with $H(f)^2 = (-\mathrm{j}\,\mathrm{sgn}\,f)^2 = -\mathrm{sgn}^2 f = -1$.)

Problem 10. Show that $x_{\mathrm{SSB}}(t) = x(t)\cos 2\pi f_c t - \hat{x}(t)\sin 2\pi f_c t$ transmits only the **upper** sideband.

Solution. Write it as a real part:

$$x_{\mathrm{SSB}} = \mathrm{Re}\left\{\big(x(t) + \mathrm{j}\hat{x}(t)\big)e^{\mathrm{j}2\pi f_c t}\right\} \quad \Rightarrow \quad x_\ell = x + \mathrm{j}\hat{x}.$$

Transform the complex envelope, using $\hat{x} \xleftrightarrow{\mathcal{F}} -\mathrm{j}\,\mathrm{sgn}(f)X(f)$:

$$X_\ell(f) = X(f) + \mathrm{j}\big(-\mathrm{j}\,\mathrm{sgn}(f)X(f)\big) = X(f)\big(1 + \mathrm{sgn}\,f\big) = \begin{cases} 2X(f), & f > 0 \\ 0, & f < 0. \end{cases}$$

So x_ℓ contains **only positive frequencies of the message**. Shifting up by f_c places this on $[f_c, f_c + W]$ — the region *above* the carrier, i.e. the upper sideband, with the mirror image at $[-f_c - W, -f_c]$ supplied by $\mathrm{Re}\{\cdot\}$. Nothing lands in $[f_c - W, f_c]$. ■

Flipping the sign of the $\hat{x}$ term gives $X_\ell = X(1 - \mathrm{sgn}\,f)$, i.e. only negative message frequencies $\Rightarrow$ lower sideband.

Problem 11. A bandpass signal has complex envelope $x_\ell(t) = \big(1 + 0.5\cos 2\pi f_m t\big)e^{\mathrm{j}\pi/4}$ with $f_m = 1$ kHz, carrier $f_c = 1$ MHz.

(a) Write $x_{bp}(t)$ explicitly.
(b) What is its bandwidth?
(c) What scheme is this?

Solution. (a) $x_{bp} = \mathrm{Re}\{x_\ell e^{\mathrm{j}2\pi f_c t}\} = \big(1 + 0.5\cos 2\pi f_m t\big)\cos\big(2\pi f_c t + \frac{\pi}{4}\big)$.

(b) x_ℓ has message content only at $\pm f_m$, so the bandwidth is $2f_m = 2$ kHz.

(c) The complex envelope is a *positive real* function times a constant phase, so the envelope is $A(t) = 1 + 0.5\cos 2\pi f_m t > 0$: this is **DSB-AM with** $\mu = 0.5$ and a single-tone message, transmitted with a 45° carrier phase. Since $A(t) > 0$ always, an envelope detector recovers it.

Problem 12. A bandpass filter has $h_{bp}(t) = 2\,\mathrm{Re}\{h_\ell(t)e^{\mathrm{j}2\pi f_c t}\}$ with $h_\ell(t) = 2W\,\mathrm{sinc}(2Wt)$.

(a) Write $h_{bp}(t)$ and $H_{bp}(f)$.
(b) A DSB-SC signal with message bandwidth $W_m < W$ passes through. Find the output complex envelope.

Solution. (a) $h_{bp}(t) = 2 \cdot 2W\,\mathrm{sinc}(2Wt)\cos 2\pi f_c t = 4W\,\mathrm{sinc}(2Wt)\cos 2\pi f_c t$, and since $H_\ell(f) = \mathrm{rect}\big(\frac{f}{2W}\big)$,

$$H_{bp}(f) = \mathrm{rect}\Big(\frac{f - f_c}{2W}\Big) + \mathrm{rect}\Big(\frac{f + f_c}{2W}\Big):$$

an ideal BPF of bandwidth $2W$ centred at f_c.

(b) By the baseband-equivalence theorem $y_\ell = x_\ell * h_\ell$, i.e. $Y_\ell = X_\ell H_\ell$. For DSB-SC, $x_\ell = Ax(t)$ with X_ℓ supported on $|f| \le W_m < W$, where $H_\ell = 1$. So

$$Y_\ell(f) = X_\ell(f) \quad \Rightarrow \quad y_\ell(t) = Ax(t),$$

the signal passes unchanged. **Moral:** the whole filtering calculation was done at baseband, with no mention of f_c — exactly why simulators and SDRs work this way.

Group D — Power and efficiency

Problem 13. A DSB-AM transmitter radiates 10 kW total with a single-tone message at $\mu = 0.7$. Find the carrier power, sideband power and efficiency. What total power would DSB-SC need to deliver the same sideband power?

Solution. Single tone $\Rightarrow P_x = \frac{1}{2}$. Then $\mu^2 P_x = 0.49 \times 0.5 = 0.245$ and

$$P_T = \frac{A^2}{2}(1 + \mu^2 P_x) = \frac{A^2}{2}(1.245) = 10 \text{ kW} \quad \Rightarrow \quad \frac{A^2}{2} = 8.03 \text{ kW (carrier)}.$$

Sideband power $= 10 - 8.03 = \boxed{1.97 \text{ kW}}$; $\eta = 1.97/10 = 19.7\%$ (check: $0.245/1.245 = 19.7\%$ ✓).

DSB-SC puts *all* its power in sidebands, so it needs only $\boxed{1.97 \text{ kW}}$ — a $5\times$ saving, at the cost of requiring a coherent receiver.

Problem 14. Show that for a message with $\max|x| = 1$, the DSB-AM efficiency cannot exceed 50%, and state the message that achieves it.

Solution. $\eta = \dfrac{\mu^2 P_x}{1 + \mu^2 P_x}$ increases monotonically in $\mu^2 P_x$, so maximise that product. The constraint is $\mu \le 1$ (from $1 + \mu x \ge 0$ with $\min x = -1$), so $\mu^2 \le 1$. And $P_x = \langle x^2 \rangle \le \max x^2 = 1$, with equality only if $|x(t)| = 1$ for all t — a **square wave** (or any ± 1-valued message). Hence

$$\mu^2 P_x \le 1 \quad \Rightarrow \quad \eta \le \frac{1}{1+1} = \boxed{50\%}.$$

For a sinusoid $P_x = \frac{1}{2}$ giving $\eta \le 1/3 = 33.3\%$; for speech, $P_x \ll 1$ (large peak-to-average ratio) and η is typically under 10%. That gap is the real reason broadcast AM is inefficient.

Problem 15. The ring modulator of Problem 4 is followed by nothing else. Its output before filtering is $x(t)c(t)$. What fraction of the output power lies in the wanted band around f_c?

Solution. $c(t) = \pm 1$ so $c^2(t) = 1$ and the total output power is $\langle x^2 c^2 \rangle = P_x$. The wanted component is $\frac{4}{\pi} x(t) \cos 2\pi f_c t$, whose power is $\left(\frac{4}{\pi}\right)^2 \frac{P_x}{2} = \frac{8}{\pi^2} P_x$.

$$\text{fraction} = \frac{8}{\pi^2} = 0.811 = \boxed{81.1\%}.$$

The remaining 18.9% sits in the harmonics at $3f_c, 5f_c, \ldots$ and is thrown away by the BPF. (Sanity check via Parseval on the Fourier series: $\frac{8}{\pi^2} \sum_{n \text{ odd}} 1/n^2 = \frac{8}{\pi^2} \cdot \frac{\pi^2}{8} = 1$ ✓.)

Night-before checklist. If you can do these six from memory you are in good shape:

1. Sketch the DSB-AM spectrum with correct heights and label the carrier's weight.
2. Derive the ring-modulator output including the $4/\pi$, and state both conditions.
3. Write the coherent detector output with phase and frequency error, and say what each does.
4. Give the envelope-detector RC inequality and evaluate it numerically.
5. Compute $x_\ell, x_I, x_Q, A, \phi$ for any given bandpass signal by the read-off method.
6. Find the image frequency and explain the role of the preselector.